

Influence of the joint treatment with granulocyte colony-stimulating factor and drugs elevating extracellular adenosine on erythropoietic recovery following 5-fluorouracil-induced haematotoxicity in mice

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Abstract: The presented data address the problem of pleiotropic effects of granulocyte colony-stimulating factor (G-CSF) and suggest the ability of drugs increasing the level of extracellular adenosine to activate erythropoiesis when given jointly with G-CSF. To demonstrate these interactions, the effects of the drugs on the recovery from erythropoietic damage induced in mice by a single dose of 5-fluorouracil (5-FU) were investigated. Elevation of extracellular adenosine and thus activation of adenosine receptors was induced by joint administration of dipyridamole (DP), a drug inhibiting the cellular uptake of adenosine, and adenosine monophosphate (AMP), an adenosine prodrug. The drugs were injected in a 4-d treatment regimen starting 2 h after 5-FU injection. Both DP + AMP and G-CSF alone induced only weak effects. However, the combination of the three drugs produced significant elevation of erythrocytes in the peripheral blood which pertained in the post-treatment period. Stimulation of proliferation of erythroid progenitor cells (BFU-E) in femoral bone marrow and increased levels of reticulocytes in the peripheral blood were observed in the course of the 4-d treatment regimen. In addition, significantly decreased mean cell haemoglobin accompanying the elevated numbers of erythrocytes in the combination-treated mice was found. This effect could be interpreted as the result of a sublethal 5-FU-induced damage to erythroid progenitor and precursor cells forced to proliferate intensively by the combination therapy. The observed additivity and synergism of G-CSF with elevated extracellular adenosine in terms of erythropoiesis is an interesting finding with potential implications in clinical practice.

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The effects of cytokines and growth factors on the haematopoietic cell amplification exhibit synergy, redundancy and pleiotropy, and suggest a complex interaction structure of the cytokine network. There is increasing evidence of the necessity to use combinations of various cytokines and growth factors to achieve higher therapeutic efficiency under myelo-

suppressive situations (1–5). Previously we have postulated a regulatory mechanism which can interact with the cytokine control of haematopoiesis and which is based on the signalling through adenosine cell receptors. It has been shown that drugs elevating extracellular adenosine and thus activating adenosine receptors can enhance the

effects of granulocyte colony-stimulating factor (G-CSF) on granulopoiesis in normal mice (6). In addition, it has been ascertained that the combination of the drugs administered to gamma-irradiated mice enhances the haematopoietic recovery not only in the granulocytic lineage, but also in the indices of erythropoiesis, monocytopenia and thrombopoiesis (7). Enhancement of the pleiotropic effects of G-CSF by the mechanisms of adenosine signalling seems to be an attractive topic which might have an impact in a clinical setting.

To address this problem, we investigated the effects of the joint administration of G-CSF and drugs elevating extracellular adenosine on the erythropoietic recovery using the model of the bone marrow damage induced by 5-fluorouracil (5-FU) in mice. 5-FU represents an antineoplastic agent which is preferentially toxic to cycling cells. Due to this action it induces myelosuppression which is manifested in all lineages of blood cells, but the 5-FU-cytotoxic activity on erythropoiesis seems to be more severe than on other lineages (8, 9). According to Hodgson *et al.* (10) this effect might be explained by a different proliferative potential in the compartments of progenitor cells, namely by a more rapid proliferation of the pre-erythroid cells as compared to granulocyte-macrophage and megakaryocyte progenitors in normal steady state.

The experimental procedure inducing the elevation of extracellular adenosine in mice utilized the previously described (6, 7) combined administration of dipyridamole (DP), a drug inhibiting the cellular uptake of adenosine (11), and adenosine monophosphate (AMP), serving as an adenosine prodrug. AMP is rapidly metabolized extracellularly to adenosine by the cell surface ectonucleotidase activity (12). Our concept is based on the expectation that both AMP, a source of exogenous adenosine, and DP, preserving endogenous and exogenous adenosine in the extracellular fluid, will act jointly in enhancing the extracellular adenosine level. Due to the action of DP, any intracellular operation of adenosine can be ruled out and the effects obtained can be attributed to the activation of extracellular adenosine receptors. In the presented studies we therefore investigated the ability of the combination of DP and AMP with G-CSF to influence erythropoiesis in mice pretreated with a single cytotoxic dose of 5-FU.

Material and methods

Animals

Male mice of the (CBA $\times$ C57BL/10)F₁ strain, 3 months old and weighing in average 25 g, were used. The mice were purchased from ANLAB Ltd.,

production facility Brno, Czech Republic. A standardized pelleted diet and HCl-treated tap water (pH 2–3) were given *ad libitum*. The mice were kept under controlled conditions (light:dark 12:12) and the temperature was maintained at 22 ± 1 °C. Animal experiments were conducted according to the principles given in the Guide for the Care and Use of Laboratory Animals issued by the Czech Society for Laboratory Animal Science.

Drugs

5-Fluorouracil (5-FU) (Sigma, St. Louis, MO, USA) was dissolved in saline and given intraperitoneally (i.p.) in a single sublethal dose of 100 mg/kg in a volume of 0.2 mL. Recombinant human G-CSF (Neupogen, Hoffman-LaRoche Ltd., Basel, Switzerland) was diluted with 5% glucose and injected subcutaneously (s.c.) at a single daily dose of 3 µg per mouse in a volume of 0.1 mL. Dipyridamole (DP) (Sigma) was dissolved in 0.4% tartaric acid and injected s.c. at a single daily dose of 2 mg per mouse in a volume of 0.4 mL. Adenosine 5'-monophosphate sodium salt from yeast (AMP) (Sigma) was dissolved in distilled water and injected i.p. at a single daily dose of 5 mg free acid per mouse in a volume of 0.2 mL. Glucose (5%), tartaric acid (0.4%), and saline were used for control injections. In the combination treatment, DP was administered 20 min before AMP, G-CSF was given 30 min after AMP. The drugs were given in a 4-d treatment regimen. The doses of drugs, their daily administration schedule and treatment regimen were based on previous experimental experience (6, 7). The first injection of drugs was given 2 h after 5-FU administration.

Haematological methods

Blood samples were taken from the ocular sinus. Numbers of erythrocytes per 1 µL of peripheral blood, mean cell volume and haematocrit values were determined by means of a Coulter counter (model ZF, Coulter Electronics, UK). Mean cell haemoglobin was measured by means of a haemoglobinometer (Coulter Electronics) using the cyanmethaemoglobin method. Blood reticulocytes were counted in blood smears stained by brilliant cresyl blue. Numbers of erythroid progenitor cells (BFU-E) in femoral bone marrow were ascertained using methylcellulose-based medium for murine progenitor assay (Institute of Haematology and Blood Transfusion, Prague, Czech Republic). The medium was composed of 1.1% (w/v) methylcellulose in Iscove's modified Dulbecco's medium, 30% (w/v) fetal bovine serum, 1% (w/v) deionized bovine serum albumin, 0.1 mM hemin, 0.1 mM 2-mercaptoetha-

nol, 4 U/mL recombinant human erythropoietin, 50 ng/mL recombinant murine stem cell factor and 5% (w/v) conditioned medium from murine interleukin-3 (mIL-3)-producing myeloma cell line (13). For determination of more mature erythroid progenitor cells, namely CFU-E, the above described medium, but without IL-3, was used. Bone marrow cells for BFU-E and CFU-E determination were cultivated for 8 and 2 d, respectively, in a humidified environment containing 5% CO₂ at 37 °C. Colonies were scored exploiting the ability of haemoglobin-containing cells to reduce 2,7-diaminofluorene (DAF) to a coloured product. 0.1 mL of DAF (Sigma), dissolved in 90% acetic acid to a concentration of 10 mg/mL, was mixed with 0.1 mL of 30% hydrogen peroxide and 10 mL of 200 mM Tris-HCl at pH 7.0. The solution was prepared in a polypropylene tube just before use, distributed evenly on top of each culture, and blue-green stained colonies were scored after 3–4 min.

Statistics

The results are presented as means $\pm$ SEM. The means of BFU-E and CFU-E are average values from individual femora from each animal. The experimental data were subjected to analysis of variance followed by detailed nonparametric analysis of differences between groups (14) using the statistical software CSS:Statistica, version 3.1 (StatSoft, Inc, Tulsa, USA, 1992). The level of significance was set at $p < 0.05$.

Results

Early effects of drug treatment

In order to investigate the early effects of G-CSF, DP + AMP, and the combination of all drugs on the

erythropoietic damage induced by 5-FU, determinations of two types of erythroid progenitor cells (BFU-E and CFU-E) in the femoral bone marrow and of blood reticulocytes were performed. As shown in Table 1, a marked drop of these indices was noted on d 2 after 5-FU injection in all the treated groups as compared to intact controls. At this time interval, two preceding injections of the drug combination, given in a 24 h interval and starting 2 h after 5-FU injection, could enhance BFU-E and CFU-E significantly, as well as reticulocyte counts compared to 5-FU-treated controls and to mice treated with G-CSF alone. Moreover, the values in combination-treated mice were found, in the case of BFU-E, to be significantly higher in comparison with mice administered G-CSF alone, in the case of CFU-E, in comparison with both G-CSF- and DP + AMP-treated mice. Concerning BFU-E, both G-CSF alone and DP + AMP induced a significant stimulatory effect at this time interval and indicate an additive action. Regarding CFU-E, apart from the combined treatment only the administration of DP + AMP was observed to produce a significant stimulatory effect in comparison with controls. On d 4 after 5-FU, i.e. 24 h after the completion of the 4-d curative regimen, the recovering counts of BFU-E exhibited the highest increase in mice treated with the combination of the drugs. Blood reticulocytes, which continued to decrease at this time interval, were not influenced by the drug treatment.

Subsequent effects of drug treatment

Selected erythroid peripheral blood parameters were determined between d 4 and d 21 after 5-FU injection, i.e. after the cessation of the curative drug treatment. As shown in Fig. 1, numbers of erythrocytes in the peripheral blood were sig-

Table 1. Numbers of BFU-E per femur, CFU-E per femur and of blood reticulocytes in intact controls and in mice subjected to curative treatment, investigated on d 2 or d 4 after 5-FU administration

Experimental groups	BFU-E per femur	CFU-E per femur	Numbers of reticulocytes per 1 μ L of peripheral blood ($\times 10^3$)
Intact controls	13908 $\pm$ 667	40032 $\pm$ 5143	178 $\pm$ 7
Day 2 after 5-FU			
Controls (tartaric acid + saline + glucose)	1079 $\pm$ 426	11096 $\pm$ 765	70 $\pm$ 9
Tartaric acid + saline + G-CSF	1287 $\pm$ 238 ^a	14752 $\pm$ 1345	48 $\pm$ 7
DP + AMP + glucose	2178 $\pm$ 585 ^a	16752 $\pm$ 1563 ^a	71 $\pm$ 8
DP + AMP + G-CSF	3394 $\pm$ 797 ^{a,B}	23482 $\pm$ 1194 ^{A,B,C}	135 $\pm$ 28 ^{A,B,C}
Day 4 after 5-FU			
Controls (tartaric acid + saline + glucose)	6588 $\pm$ 1504	ND	5.5 $\pm$ 4.9
Tartaric acid + saline + G-CSF	9176 $\pm$ 1294	ND	4.0 $\pm$ 2.1
DP + AMP + glucose	8449 $\pm$ 1263	ND	6.1 $\pm$ 2.7
DP + SMP + G-CSF	11471 $\pm$ 917 ^a	ND	4.4 $\pm$ 3.9

Data are given as means $\pm$ SEM for 5 mice per group. ^A $p < 0.01$ and ^a $p < 0.05$, respectively, in comparison with 5-FU-treated controls. ^B $p < 0.01$, in comparison with G-CSF-treated mice. ^C $p < 0.01$, in comparison with DP + AMP-treated mice. Values that are not superscripted are not significantly different from those of the pertinent experiment or control groups. ND, not done.

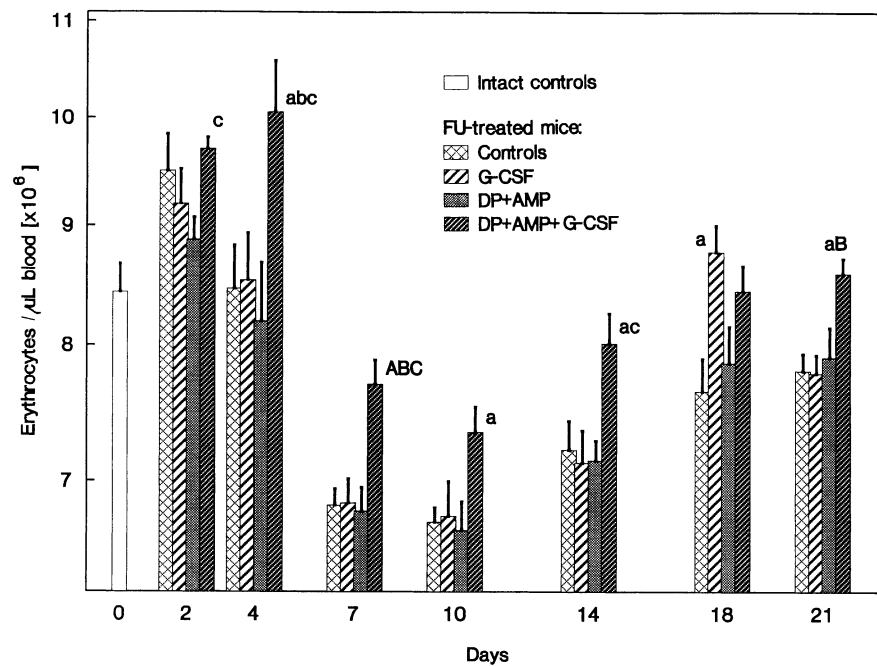


Fig. 1. Numbers of erythrocytes per 1 μ L of the peripheral blood in differently treated groups of mice. Data are given as means $\pm$ SEM for 10 mice per group. Days, days after 5-FU administration. Day 0, value in intact control mice. A and a, $p < 0.01$ and $p < 0.05$, respectively, in comparison with 5-FU-treated controls. B and b, $p < 0.01$ and $p < 0.05$, respectively, in comparison with G-CSF-treated mice. C and c, $p < 0.01$ and $p < 0.05$, respectively, in comparison with DP+AMP-treated mice. Values that are not marked by A, a, B, b, C or c are not significantly different from those of the pertinent experimental or control groups.

nificantly increased in mice treated with the combination of the drugs in comparison with 5-FU-treated controls on d 4, 7, 10, 14, and 21. No significant effects of DP+AMP or G-CSF alone were found with the exception of d 18, when numbers of erythrocytes in mice given only

G-CSF were significantly higher compared to 5-FU-treated controls. Because of non-existing differences in the mean cell volume between differently treated groups, the haematocrit values behaved similarly as the numbers of erythrocytes (data not given).

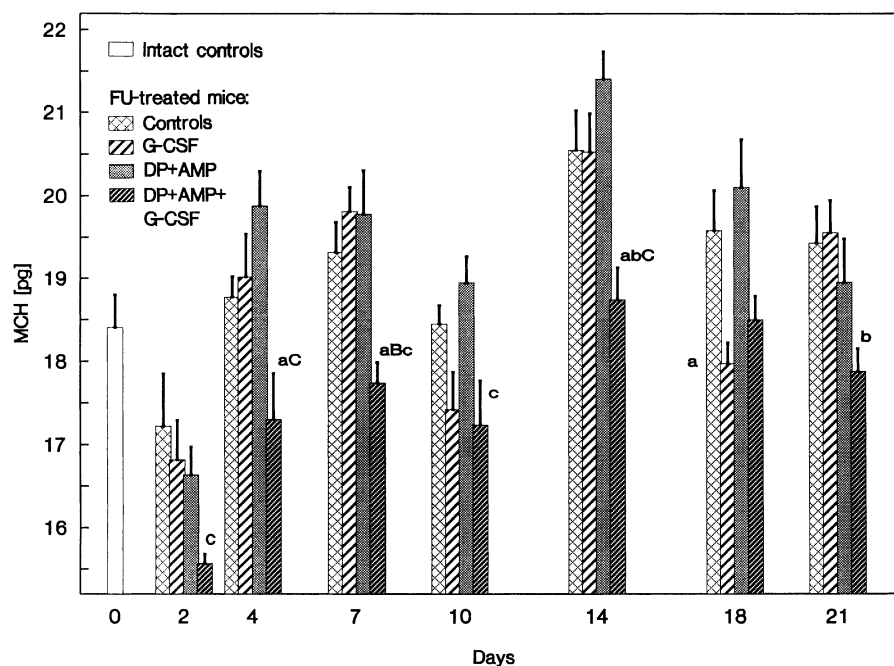


Fig. 2. Mean cell haemoglobin (MCH) in differently treated groups of mice. For other explanations, see Fig. 1.

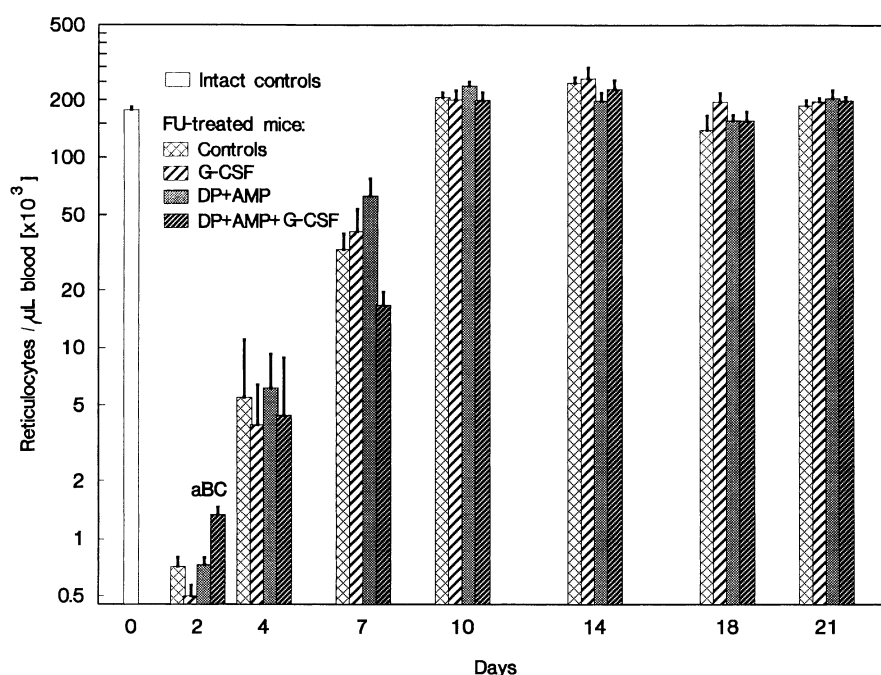


Fig. 3. Numbers of reticulocytes per 1 μL of the peripheral blood in differently treated groups of mice. For other explanations, see Fig. 1.

The mean cell haemoglobin (MCH) values (Fig. 2) in mice treated with the combination of the drugs exhibited a decrease which attained statistical significance on d 4, 7 and 14 compared to 5-FU-treated controls. No significant effects of DP+AMP or G-CSF alone were noted with the exception of d 18, when the MCH values in mice given only G-CSF were significantly decreased compared to 5-FU-treated controls.

Blood reticulocytes were not significantly affected by the action of the drugs between d 4 and d 21 after 5-FU administration (Fig. 3). This suggests an undisturbed operation of recovery mechanisms after the cessation of the curative regimen.

Discussion

The effects of a single dose of 5-FU on erythropoiesis in control mice observed in our experiments are comparable to those reported earlier (9, 15). Damage to bone marrow cells is indicated by high reduction of the erythroid progenitor cells on d 2 after 5-FU administration. While peripheral reticulocytes are decreased on d 4 to very low levels and start to rise on d 7, peripheral erythrocytes exhibit a continuous decline, reach a nadir on d 10, then start to increase and approach normal levels at the end of the third week.

The purpose of this study was to compare the erythropoietic action of G-CSF alone, of DP+AMP, and of the combination of all the drugs administered in a 4-d curative regimen shortly

after 5-FU. The statistical analysis of the differences among the compared groups revealed an expressive stimulatory effect of the combination therapy. Significant effects of this treatment were shown not only when compared to controls, but also in several indices and intervals in comparison to groups treated with G-CSF or DP+AMP. The single components of the drug combination scarcely induced significant stimulatory effects, which indicates their moderate action. Enhanced levels of peripheral erythrocytes occurring on d 4 after 5-FU in mice treated with the combination of the drugs should be considered as a result of increased cell production. Due to the lifespan of erythrocytes in the peripheral blood, their increased numbers are maintained during the whole period of observation after cessation of the treatment. Theoretically, determinants of cell amplification can be cycling activity increasing the size of the progenitor cell pool, the cell cycle time of dividing progenitor and precursor cells, the transit time of dividing cells and the level of apoptosis (4, 16). Our data do not make it possible to ascertain the role of these possible mechanisms, but give evidence of cell amplification in the pool of erythroid progenitor cells in the bone marrow. The contribution of splenic erythropoiesis to cell production cannot be excluded, but the determination of spleen weight on d 2 and d 4 after 5-FU (data not given) did not indicate the activation of this extramedullary organ of erythropoiesis. Rich (9) has shown in 5-FU-treated mice that, in contrast to the bone marrow, the spleen

cellularity remained within the control range throughout the whole period of marrow depletion and recovery. It is uncertain whether the elevation of reticulocytes induced by the combination therapy at d 2 after 5-FU is mediated by new production of cells or by an earlier release of pre-existing reticulocytes from the haematopoietic tissue. The lack of the action of the treatment on reticulocytes on d 4 can be due either to the exhaustion of reserves, or to the operation of regulatory feedback mechanisms induced by increased haematocrit values. Changes in haematocrit itself are an important mechanism in the homeostasis of red cell mass and an elevated haematocrit has an inhibitory influence on erythropoiesis (17). Such an effect could interfere with the mechanisms of positive control which are activated by the 5-FU-induced damage and by the action of the drugs. The lack of differences in the numbers of reticulocytes among the groups of mice studied in the time interval from day 4 up to day 21 strongly suggests that stimulatory effects on progenitor cell proliferation evoked by the combined treatment are limited to early intervals following drug administration. The brief duration of the stimulatory effects is clearly related to the short half-life of exogenous adenosine in the extracellular fluid (18).

An interesting phenomenon observed in our experiments is the decrease in MCH values accompanying the increased counts of peripheral erythrocytes in animals subjected to the combination therapy. The possible explanation can be sublethal damage to the erythroid cells induced by 5-FU. This cytotoxic drug does not only inhibit thymidylate synthase activity and thus DNA synthesis, but it is probably also incorporated into RNA of non-S-phase cells of the erythron, and thus the haemoglobin synthesis may be damaged (8, 19, 20). When, subsequently, such damaged cells are induced into a more intensive proliferation, hypochromic cells can be produced.

Both G-CSF and extracellular adenosine elevated by the administration of dipyrindamole and AMP exert their biological effects by binding to specific receptors on the cell surface. Adenosine acts as a paracrine regulator in many tissues through the activation of G protein-coupled receptors which initiate signal transduction mechanisms. These receptors have been categorized as A₁, A_{2A}, A_{2B}, and A₃ subtypes. The A₁ and A₃ receptors inhibit adenylate cyclase leading to a decrease in cellular cAMP levels, while the A₂ receptors activate adenylate cyclase and induce an increase in cellular cAMP levels (21). Many data suggest that signalling through adenosine receptors is an important regulatory mechanism modulating cell cycle, differ-

entiation, viability and apoptosis of cells, and enhancement of cell proliferation by extracellular adenosine has been repeatedly reported (for reviews see ref. 22–24). There exists evidence of synergistic interactions between the effects of adenosine receptor signalling and the effects of polypeptidic growth factors. For example, Huang and coworkers (25) have shown in various types of cells *in vitro* that adenosine and epidermal growth factor have very small mitogenic effects when tested alone but synergistically stimulate DNA synthesis when added together to the cells. Our results give another example of the additivity or synergism of different signals which activated alone are insufficient to induce an expressive effect. Mechanisms of these interactions remain to be clarified. Some data suggest that the G-protein/adenylate cyclase/cAMP system may have a role in G-CSF signal transduction for proliferation of immature myeloid cells (26). Relevant to our findings can be data indicating that erythropoiesis, as assayed by erythroid colony formation *in vitro*, is enhanced by adenine nucleotides and agents which are capable of raising intracellular cAMP levels (27). Moreover, activation of adenosine receptors of the A₂ subtype was found to increase the production of erythropoietin in mice (28) and rats (29). Thus, direct cellular as well as indirectly mediated effects might participate in the action of extracellular adenosine on erythropoiesis. Such effects could act jointly with the pleiotropic effects of G-CSF, including its ability to stimulate erythropoietic functions (30, 31). Considering the observations reported here and our earlier findings suggesting stimulatory effects of extracellular adenosine on the formation of endogenous haematopoietic spleen colonies in gamma-irradiated mice (32) and the additivity of these effects with the effects of G-CSF (33), it cannot be excluded that these interactions are realized at the level of the early haematopoietic progenitor cells.

It can be summarized that our results give further experimental evidence of the role of extracellular adenosine receptors in the regulation of haematopoietic functions and confirm the ability of this regulatory pathway to interact with the effects of haematopoietic growth factors. The interaction of G-CSF (generally regarded as a neutrophil-specific growth factor) with extracellular adenosine in terms of erythropoiesis is surprising and might have significant pharmacologic implications in curative strategies. The introduction of the concept of adenosine receptor signalling into haematology seems to be a promising, but yet underestimated possibility.

Previously published findings (6, 7) as well as the results presented in this paper suggest the possibility

of using the enhancement of extracellular adenosine level for management of human subjects given 5-FU as anticancer agent. Before doing so it will be necessary to test whether the application of DP + AMP does not also stimulate the proliferation of tumour cells. The authors plan to evaluate the effects of the drugs on tumour-bearing mice.

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